

Set Theory:

Preliminaries + Motivation:

Def. (Experiment): An activity or process whose outcome is: i) of interest, ii) subject to uncertainty.

Classic Examples:

- Flipping a Coin
- Rolling a Die
- Dealing a hand of 5 cards

An experiment can have several possible outcomes,

e.g. a coin may land heads-up or tails-up.

a die can show 1, 2, 3, 4, 5, or 6

Def. (Sample Space): Given an experiment, the sample space S is the set of all possible outcomes.

Examples:

- Coin Flip, $S = \{H, T\}$
- Die Roll, $S = \{1, 2, 3, 4, 5, 6\}$
- Flip 2 Coins, $S = \{(H, H), (H, T), (T, H), (T, T)\}$

1st
coin

}

2nd
coin

A key word in the above is **set**. If we want to study different possible outcomes, we will need to understand how to work with sets.

Introduction to Set Theory:

Start by formally defining what a set is!

Def. (Set): A **set**, here denoted **A** , is any (possibly infinite) collection of different mathematical objects. It can contain: numbers, symbols, or any other items.

Note: We write sets by specifying their items between curly brackets, e.g. the set $A = \{1, 2, 3\}$ contains the three items '1', '2', and '3'.

Note: Sets are uniquely defined by the items they contain. So order does not matter. The two sets $A = \{H, T\}$ and $B = \{T, H\}$ are equivalent. We would write this as " $A = B$ ".

Note: The number of items in A (the **cardinality** of A), is denoted **$|A|$**

Note: Sets cannot contain duplicate items, so $A = \{1, 1, 2, 3\}$ is not a set.

There is one special set which contains no items whatsoever.

Def. (Empty Set): The **empty set**, **$\emptyset$** , is defined as **$\emptyset = \{\}$** .

Relationships Between Sets & Items

Given a set and an item, we may first ask:

"is this item in the set or not?"

Let x denote some item, and A be some set.

We write:

- $x \in A$ (read: " x in A ") if A contains item x
- $x \notin A$ (read: " x not in A ") if A does not contain item x

Notice that exactly one of these two must be true, regardless of what x and A are.

Ex.: $A = \{1, 2, 3\}$

- $x = 1 ? \Rightarrow x \in A$
 - $x = 4 ? \Rightarrow x \notin A$
 - $x = 3.0001 ? \Rightarrow x \notin A$
-

This has only told us about the relationship between a single item and a single set. However, in many interesting settings we have several different sets - and would like to study the relationship between them.

Goal: Find a way of seeing if one set is "contained in" another.

Def. (Subset): Given two sets A and B , we say that A is a subset of B , denoted $A \subset B$ (or $A \subseteq B$) whenever every item in A is also in B .

Mathematically: $(\underbrace{\forall x \in A}_{\text{"for all } x \text{ in } A}}, \text{ we have } x \in B) \Rightarrow \underbrace{A \subset B}_{\text{"implies that"}}$

Note: There are other commonly used notations/concepts for set-to-set relationships:

- If A is not a subset of B , then we will write $A \not\subset B$.
- If A is a subset of B , but A is not equal to B (i.e. B contains at least one item not in A), then we write $A \subsetneq B$, read: " A is a strict subset of B "

Examples: $B = \{1, 2, 3, 4, 5, 6\}$

- $A = \{2, 4, 6\} \rightsquigarrow A \subset B$ (moreover $A \subsetneq B$)
- $A = \{2\} \rightsquigarrow A \subset B$ ($\neq A \subsetneq B$)
- $A = \{2, 7\} \rightsquigarrow A \not\subset B$, because $7 \notin B$
- $A = 2 \rightsquigarrow A$ is not a set, so A cannot be a subset of B .

Q: How can we show that for some sets A, B , we have $A \subset B$?

A: Using the definition, we need to show that any item in A must be in B . This implies that $A \subset B$.

Clearly, $A \subset B$ does not mean that A equals B .

(See examples above) B might contain some other items that are not in A . However, subsets can be used to show two sets are equal.

Prop. (Equality of Sets): Given two sets A and B , if both $A \subset B$ and $B \subset A$, then $A = B$.

This is by far the most common way to prove that two sets are equal. First show that any item in A must be in B , then show that any item in B must be in A .

Set Operations:

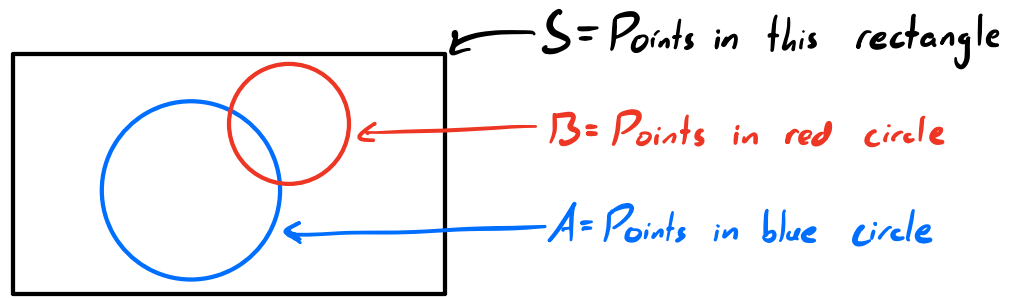
There are three main operations we can do on one or more sets that will yield a different set. They are as follows.

Def. (Union, Intersection, Complement): Let A , B , and S be three sets, such that both $A \subset S$ and $B \subset S$.

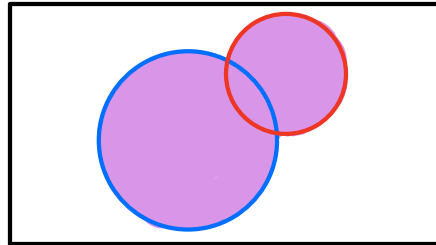
- The **union of A and B** , denoted $A \cup B$ (read: "A or B"), is a set containing all items in A and all items in B .
Formally, $(x \in A \text{ or } x \in B) \Rightarrow x \in (A \cup B)$
- The **intersection of A and B** , denoted $A \cap B$ (read: "A and B"), is a set containing all items that are in both of A , B .
Formally, $(x \in A \text{ and } x \in B) \Rightarrow x \in (A \cap B)$
- The **complement of A in S** , denoted A^c (read: "A complement"), is a set containing all items in S except those in A .
Formally, $(x \in S \text{ and } x \notin A) \Rightarrow x \in A^c$

Note: The set S is usually inferred from context, and so we do not always specify "the complement of A in S ", instead saying only "the complement of A "

Visual of Set Operations:

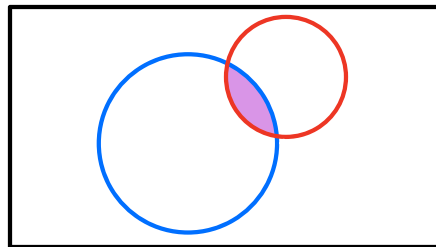


$A \cup B$:



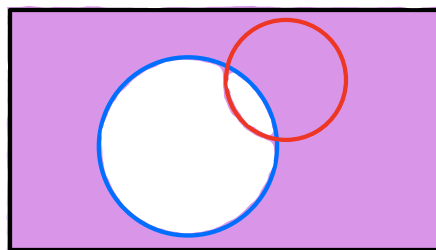
$\leadsto$ Points in either A or B

$A \cap B$:



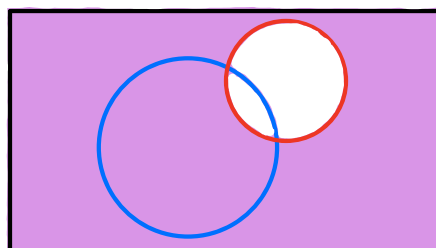
$\leadsto$ Points in both A and B

A^c :



$\leadsto$ Points in S outside of A

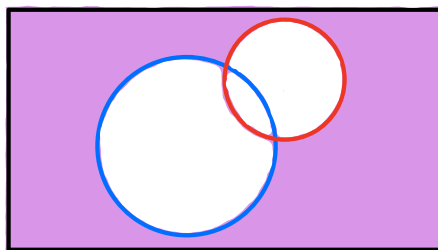
B^c :



$\leadsto$ Points in S outside of B

These set operations can be used in sequence. For example, the set $(A \cup B)^c$ would look like:

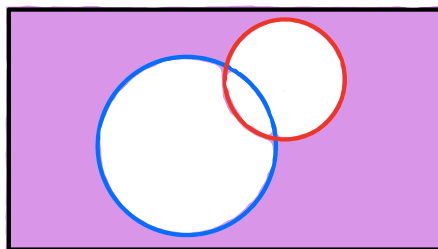
$(A \cup B)^c$:



~ Points in S outside of those in either A or B

What about a different combination? Consider the set $A^c \cap B^c$. It looks like:

$A^c \cap B^c$:



~ Points both in S outside A and in S outside B

→ The same two sets of points!

Lemma (De Morgan's Law): If A and B are both subsets of S , then: $(A \cup B)^c = A^c \cap B^c$.

In fact, we can extend this result to when we have more than two sets!

Lemma (De Morgan's Law w/ Multiple Sets): If $A_1, \dots, A_n$ are all subsets of S , then: $\underbrace{(A_1 \cup A_2 \cup A_3 \dots \cup A_n)^c}_{\bigcup_{i=1}^n A_i} = \underbrace{A_1^c \cap A_2^c \cap A_3^c \dots \cap A_n^c}_{\bigcap_{i=1}^n A_i^c}$

Examples:

Let $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, and $B = \{1, 3, 5\}$

Then:

- $A \cup B = \{1, 2, 3\} \cup \{1, 3, 5\} = \{1, 2, 3, 5\}$
 - $A \cap B = \{1, 2, 3\} \cap \{1, 3, 5\} = \{1, 3\}$
 - $A^c = \{4, 5, 6\}$, $B^c = \{2, 4, 6\}$
 - $(A \cup B)^c = \{1, 2, 3, 5\}^c = \{4, 6\}$
 - $A^c \cap B^c = \{4, 5, 6\} \cap \{2, 4, 6\} = \{4, 6\}$
 - $A \cap A^c = \{1, 2, 3\} \cap \{4, 5, 6\} = \emptyset$
-

It is entirely possible for the intersection of two sets to be empty - a simple example is $A \cap A^c$, regardless of what A contains. This property has a special name.

Def. (Disjoint): If A and B are two sets for which

$A \cap B = \emptyset$, then we say that A and B are disjoint.

Ex: Disjoint Sets

- $A, B = A^c$
- $A = \{\text{even integers}\}$, $B = \{\text{odd integers}\}$
- $A = \{\text{negative integers}\}$, $B = \{\text{positive integers}\}$

A (potential) limitation of this definition is that it only applies to pairs of sets - sometimes we have more.

Given some number of sets, here are two properties they might have.

Def. (Mutually Exclusive, Collectively Exhaustive): Let $A_1, A_2, \dots, A_n$ be some sets, such that $A_1 \subset S, A_2 \subset S, \dots$ and $A_n \subset S$.

- If for any $1 \leq i < j \leq n$, $A_i \cap A_j = \emptyset$, then $A_1, \dots, A_n$ are said to be **mutually exclusive**.

- If $\bigcup_{i=1}^n A_i = S$, then $A_1, \dots, A_n$ are said to be **collectively exhaustive**.

Note: Intuitively, "mutually exclusive" $\approx$ "none of them overlap", and "collectively exhaustive" $\approx$ "every item (in S) is in at least one"

Ex.: $S = \{1, 2, 3, 4, 5, 6\}$

- $A_1 = \{1, 2, 3\}, A_2 = \{4, 5\}, A_3 = \{6\} \rightsquigarrow$ ^(mutually exclusive) M.E. and ^(collectively exhaustive) C.E.
- $A_1 = \{1, 2\}, A_2 = \{2, 3, 4\}, A_3 = \{5, 6\} \rightsquigarrow$ not M.E., but C.E.
- $A_1 = \{1, 3, 5\}, A_2 = \{2, 4\} \rightsquigarrow$ M.E., but not C.E.

Example: Sets

Prove that if A, B are subsets of S , then $A = (A \cap B) \cup (A \cap B^c)$

Part 1) Show that $A \subset (A \cap B) \cup (A \cap B^c)$

Let $x \in A$ be some item. It must be true that either: i) $x \in B$, or ii) $x \notin B$. Consider each case:

i) $x \in B$. Then $x \in (A \cap B)$, since x is in both A and B .

Since $x \in (A \cap B)$ then $x \in \underline{(A \cap B) \cup (A \cap B^c)}$, by def'n of union

① So: $x \in A$ and $x \in B$ implies that $x \in \underline{(A \cap B) \cup (A \cap B^c)}$.

ii) $x \notin B$. Then, it must be that $x \in B^c$.

Since $x \in A$ and $x \in B^c$, we will have $x \in (A \cap B^c)$.

Therefore, $x \in (A \cap B^c) \cup (A \cap B)$, again by def'n of union.

② So: $x \in A$ and $x \notin B$ imply that $x \in \underline{(A \cap B) \cup (A \cap B^c)}$

Putting together ① and ② gives: $x \in A$ implies $x \in (A \cap B) \cup (A \cap B^c)$.

So by def'n of subset, we have that $A \subset \underline{(A \cap B) \cup (A \cap B^c)}$. ✓

Part 2) Show that $(A \cap B) \cup (A \cap B^c) \subset A$

Let $x \in (A \cap B) \cup (A \cap B^c)$. Then we must have at least one of either: i) $x \in (A \cap B)$ or ii) $x \in (A \cap B^c)$. Consider each case:

i) $x \in A \cap B$. This implies that x must be in both A and B .

Hence x must be in A .

① So: $x \in A \cap B$ implies that $x \in A$.

ii) $x \in A \cap B^c$. Hence x must be in A .

①② So: $x \in A \cap B^c$ implies that $x \in A$.

Putting together ① and ①② gives: $x \in (A \cap B) \cup (A \cap B^c)$
implies $x \in A$.

Hence, by def'n of subset, we have $\underline{(A \cap B) \cup (A \cap B^c)} \subset A$. ✓

Put together both parts to conclude that $A = \underline{(A \cap B) \cup (A \cap B^c)}$ ▯

If Time: Russell's Paradox [Not on HW/Tests]

We have not restricted the items that sets can contain in any way. So - can sets contain other sets?

In principle, yes! Here is one such set:

$$A = \{ \underbrace{\{1\}}, \{1, 2\}, \{1, 2, 3\} \}$$

↳ Note that this is different from the number "1".
It is instead the set containing only the number "1".

It satisfies the requirement in our definition - a collection of different mathematical objects.

An observation: Some sets may contain themselves

Ex.: $A =$ the set of all sets

i.e. if B is a set (any set), then $B \in A$

But A is itself a set, so hence $A \in A$!

Notice that there is nothing wrong about how we defined A ...

Consider the following set:

$A =$ the set of all sets that do not contain themselves.

i.e. If S is a set, and $S \notin S$, then $S \in A$

Q: Does this set A contain itself?

A: Let's try and work out the possibilities.

- If A does not contain itself, then $A \notin A$
- But since A is a set that does not contain itself, it satisfies the condition for being in A .
- So A must contain itself, $A \in A$

~> Reached a contradiction, so it cannot be that $A \notin A$

- If A does contain itself, then $A \in A$
- But since $A \in A$, it must be that A satisfies the condition for being in A
- So A must be a set that does not contain itself, $A \notin A$.

~> Another contradiction, so it can't be that $A \in A$

Conclusion: It's impossible for A to contain itself, and it is impossible for A to not contain itself.

Hence, A neither contains itself, nor doesn't contain itself...?

~> Russell's Paradox